

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

ServiceNow Implementation — Flow Designer based IT Procurement Automation

Project Detail	Value
Team ID	SWTID-2026-9532
Team Members	Vishal M; Hari Prasath M; Jenusha Fredrin J; Amudhan A S
Project Platform	ServiceNow
Core Technology	Service Catalog + Flow Designer
Project Behavior	Approval-based automation
Documentation Date	01 October 2026

1. Introduction

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Project objective: Automate IT hardware procurement for a Standard Laptop request so that, after the required approval condition is satisfied, a Catalog Task is created and assigned to the Hardware team. The solution integrates Service Catalog, Requested Items, Approvals, Flow Designer and Catalog Tasks.

Team Members and Roles

Team Member	Role
Vishal M	Team Lead
Hari Prasath M	Member
Jenusha Fredrin J	Member
Amudhan A S	Member

2. Project Overview

The solution is designed to reduce manual intervention in standard laptop procurement. A requester places a Standard Laptop order through the Service Catalog, the request generates a Requested Item and approval record, and the configured Flow Designer solution is associated with the catalog item.

Key features

- Standard Laptop catalog item under Hardware.
- Flow: **Standard laptop task**.
- Service Catalog / Catalog Item Requested trigger.
- Create Catalog Task using the trigger Requested Item.
- Short description: **laptop need to configured**.
- Assignment group: **Hardware**.
- Process Engine association with Standard Laptop.
- Request, approval, flow and task records can be verified in ServiceNow.

3. Architecture

This project is implemented inside ServiceNow rather than as a standalone React/Node/MongoDB application. The template's architectural layers are therefore mapped to the ServiceNow platform.

Frontend

ServiceNow web interface provides Service Catalog, Standard Laptop ordering, request/approval views, Requested Items, Flow Designer and Catalog Task views.

Backend / Automation

Flow Designer provides the automation layer. The flow uses the Catalog Task [sc_task] table, trigger Requested Item, task description/short description and Hardware assignment.

Data stores

Relevant records include sc_request, sc_req_item, sysapproval_approver and sc_task.

4. Setup Instructions

Prerequisites

Windows 10/11, macOS or Linux; Chrome, Edge or Firefox; stable internet; active ServiceNow instance/access; Service Catalog and Flow Designer enabled; Standard Laptop catalog item.

Configuration

1. Open Flow Designer / Workflow Studio.
2. Configure **Standard laptop task**.
3. Use Service Catalog / Catalog Item Requested trigger.
4. Add Create Catalog Task.
5. Use Catalog Task [sc_task].
6. Set short description and Hardware assignment.
7. Associate the flow with Standard Laptop under Process Engine.
8. Place a test order and complete approval.
9. Verify Requested Item and Catalog Task.

Approval-gating note: UAT identified that a task can be created before the required approval condition. An explicit approval condition/wait is required before task creation so pending/rejected requests do not generate fulfilment tasks.

5. Folder / Logical Structure

Logical area	Contents
Service Catalog	Hardware → Standard Laptop
Request layer	Request → Requested Item
Approval	Approver record and state
Automation	Flow Designer → Standard laptop task
Task layer	Catalog Task [sc_task] → Hardware
Validation	Request, RITM, flow execution and task views

6. Running the Application

The project is run through the ServiceNow instance. Open Service Catalog → Hardware → Standard Laptop → Order Now, submit the request, complete approval, then inspect the Requested Item and Catalog Tasks.

Recorded test order: **REQ0010004**, placed 01 October 2026 at 01:54:01. Approval was recorded as Approved. The documented resulting task is SCTASK0010004.

Figure 1 — Order Status / Request Submission

The screenshot displays the ServiceNow user interface for viewing an order status. The top navigation bar shows the 'Order Status: REQ0010004' with a star icon. Below the navigation bar, a green confirmation banner states 'Thank you, your request has been submitted'. The main content area provides details for the order placed on 2026-10-01 at 01:54:01, with request number REQ0010004 and an estimated delivery date of 2026-10-01. A table lists the items ordered:

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-10-01		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

At the bottom of the page, there are buttons for 'Back to Catalog', 'Continue Shopping', and 'Home'. The left sidebar shows the 'service cat' search results and a list of favorites and all results, including 'Self-Service', 'Service Catalog', 'Catalog Builder', 'Request Overview', 'Catalogs', 'Catalog', 'Open Records', 'Requests', 'Items', and 'Tasks'.

Figure 2 — Request Approval

The screenshot shows the ServiceNow interface for a specific request, REQ0010004. The left sidebar contains navigation options like 'Service Catalog', 'Catalog Builder', and 'Open Records'. The main area displays the request details, including 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. Below this, there are tabs for 'Requested Items (1)', 'Approvers (1)', and 'Recurring Prices (1)'. The 'Approvers (1)' tab is active, showing a table with one approval record.

State	Approver	Comments	Created
Approved	Eric Schroeder		2026-10-01 01:54:02

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 2:24 PM on 10/1/2026.

REQ0010004 approval record showing the documented Approved state.

Figure 3 — Requested Items Overview

The screenshot displays the 'Requested Items View: Self Service' interface. The left sidebar shows navigation options like 'Requested Items', 'Watched Requested Items', and 'Help Requests'. The main area features a table listing requested items with columns for 'Number', 'Catalog', 'Item', 'Approval', 'Quantity', and 'Stage'.

Number	Catalog	Item	Approval	Quantity	Stage
RITM0010005	(empty)	Standard Laptop	Requested	1	
RITM0010004	(empty)	Standard Laptop	Approved	1	
RITM0010003	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0010002	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0010001	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0000005	(empty)	Standard Laptop	Requested	1	
RITM0000004	(empty)	Standard Laptop	Requested	1	
RITM0000003	(empty)	Standard Laptop	Requested	1	
RITM0000002	(empty)	Samsung Galaxy S7	Requested	1	

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 3:20 PM on 10/1/2026.

Requested Items list showing the recorded approval states, including RITM0010004 as Approved and RITM0010005 as Requested.

7. API Documentation

No custom REST or external backend API endpoints are documented. The implementation uses native ServiceNow records and Flow Designer actions.

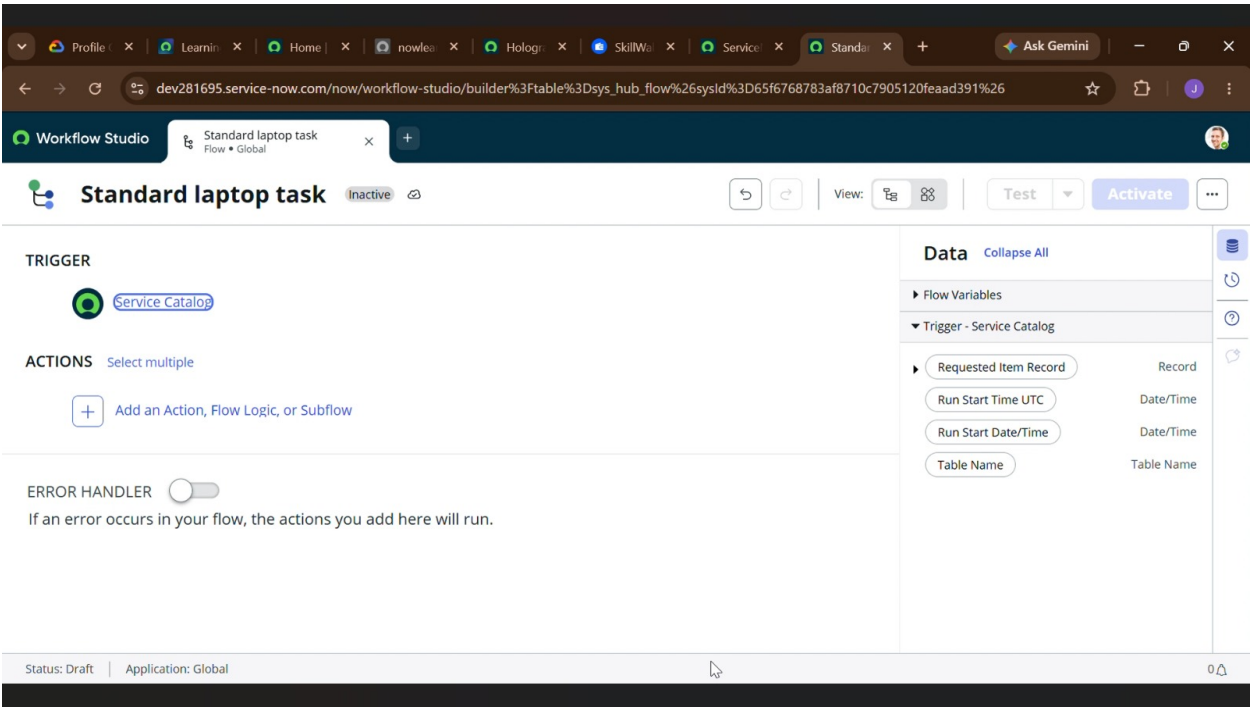
Interface / record	Method	Purpose
Service Catalog	UI action	Submit Standard Laptop request
Approval record	Record update	Approve or reject request
Flow Designer	Automation	Process catalog-item request
sc_task	Record creation	Create Hardware fulfilment task

8. Authentication

The project uses the ServiceNow instance's configured access, roles and permissions. Credentials are not included in the report.

9. User Interface & Configuration Evidence

Figure 4 — Standard Laptop Flow Overview



Workflow Studio view showing the Standard laptop task flow, its Catalog Item Requested trigger and Create Catalog Task action.

Figure 5 • Create Catalog Task Configuration

The screenshot displays the 'Standard laptop task' configuration interface. The 'Table Name' is set to 'Catalog Task [sc_task]'. The 'Requested Item' is configured with 'Trigger -> Requested Item'. The 'Short Description' is 'laptop need to configured'. The 'Description' field also contains 'laptop need to configured'. The 'Assignment group' is set to 'Hardware'. The 'Approval' field is set to 'Approved'. The 'Template Catalog Item' is set to 'Select Template Catalog Item'. The 'Data' panel on the right shows flow variables including 'Requested Item Record', 'Run Start Time UTC', 'Run Start Date/Time', and 'Table Name'.

Create Catalog Task configuration showing sc_task, Requested Item input, task description, Hardware assignment and approval field.

Figure 6 — Standard Laptop Process Engine

The screenshot shows the 'Standard Laptop' catalog item configuration page. The 'Process Engine' tab is active, showing the 'Flow' field set to 'Standard laptop task'. The 'Workflow' and 'Execution Plan' fields are empty. The 'Item Details' tab shows the 'Category' set to 'Hardware', 'State' as '-- None --', 'Checked out' as '-- None --', and 'Owner' as an empty field. The 'Secondary owners' field is also empty.

Standard Laptop catalog item Process Engine configuration showing the associated Standard laptop task flow.

Figure 7 · Standard Laptop Catalog Item

Build and modify items faster with the improved [Catalog Builder](#).

Name: Standard Laptop

Application: Global

Active: ☒

Roles: [Edit](#)

Category: Hardware

State: -- None --

Checked out: -- None --

Owner: [Select](#)

Secondary owners: [Add](#)

Fulfillment automation level: Unspecified

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description: Lenovo - Carbon x1

Description: [Add](#)

ServiceNow Catalog Item record for Standard Laptop under the Hardware category.

Figure 8 — Flow Executions

Flow executions ²

Last refreshed just now.

Name	State	Runtime	Created by	Created	Sys ID
Standard laptop task	Waiting	187	admin	2026-10-01 02:49:22	53afb9b7c2f031011ca98fa5ed323f6
Standard laptop task	Waiting	1,030	admin	2026-10-01 01:54:25	ea1333dbd3ab03106228dfdaf06cd43b

Showing 1-2 of 2

1 rows per page

Flow Executions view showing recorded Standard laptop task executions and their Waiting state.

10. Testing

The latest documented UAT state contains 8 test cases: 6 Pass, 1 Fail and 1 Not Tested.

Test Case	Scenario	Result
TC-001	Submit Standard Laptop request	PASS
TC-002	Verify Requested Item creation	PASS
TC-003	Verify approval record/state	PASS
TC-004	Verify flow association	PASS

Test Case	Scenario	Result
TC-005	Verify Catalog Task exists for approved request	PASS
TC-006	Verify short description and Hardware assignment	PASS
TC-007	Verify no task while approval is pending	FAIL
TC-008	Rejected-request scenario	NOT TESTED

Open defect

Bug ID	Description	Severity	Status
BG-001	Catalog Task is created before the required approval condition; a pending request was observed with a Catalog Task and a rejected request was also reported to have a task.	Severity 2	Open

Required correction: introduce an explicit approval condition/wait before Create Catalog Task so only appropriately approved requests reach task creation.

11. Screenshots / Demo

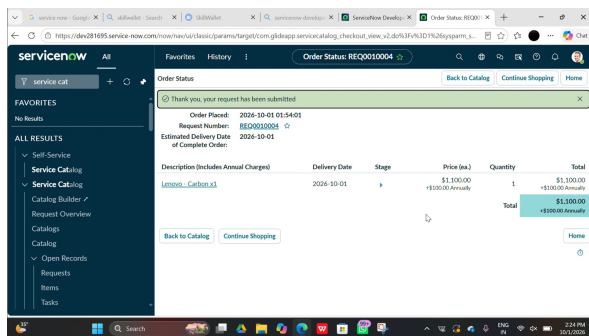


Figure 11.1 — Standard Laptop order status / REQ0010004

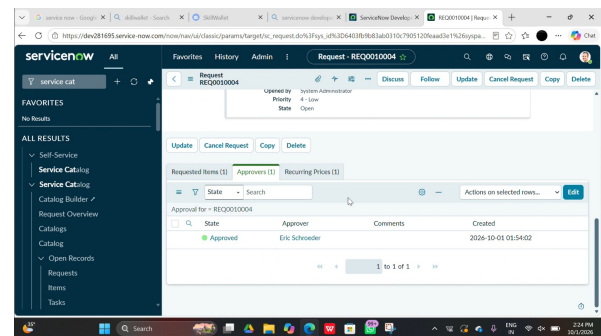


Figure 11.2 — REQ0010004 approval record

Additional non-duplicate project evidence continues on the following pages.

Requested Items View: Self Service

Number Search

Actions on selected rows... New

All > Request Requested for = System Administrator > Active = true

Number	Catalog	Item	Approval	Quantity	Stage
RITM0010005	(empty)	Standard Laptop	Requested	1	
RITM0010004	(empty)	Standard Laptop	Approved	1	
RITM0010003	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0010002	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0010001	(empty)	Standard Laptop	Not Yet Requested	1	
RITM0000005	(empty)	Standard Laptop	Requested	1	
RITM0000004	(empty)	Standard Laptop	Requested	1	
RITM0000003	(empty)	Standard Laptop	Requested	1	
RITM0000002	(empty)	Samsung Galaxy S7	Requested	1	

1 to 10 of 10

Figure 11.3 — Requested Items and approval states

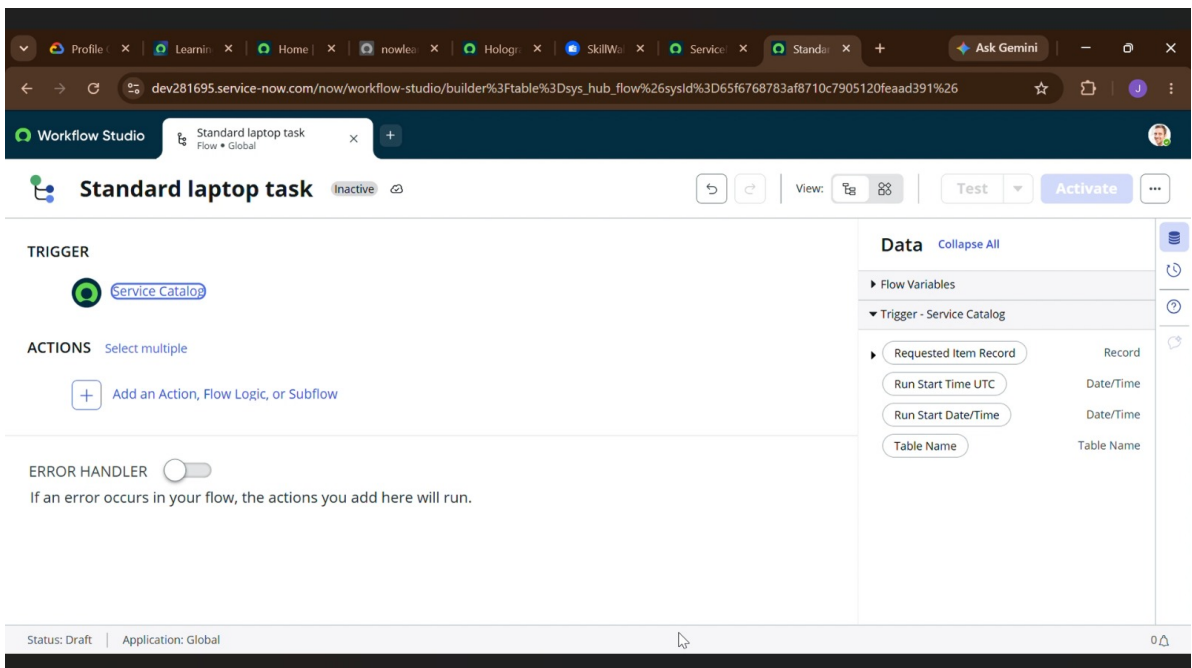


Figure 11.4 — Standard laptop task flow overview

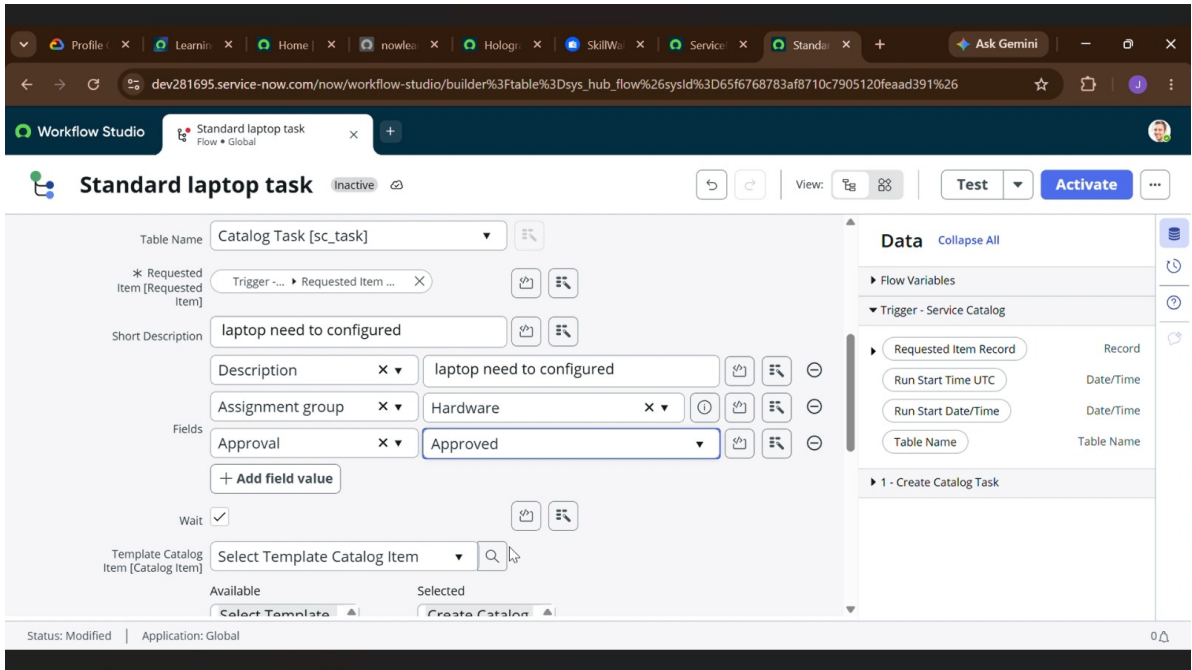


Figure 11.5 — Create Catalog Task configuration

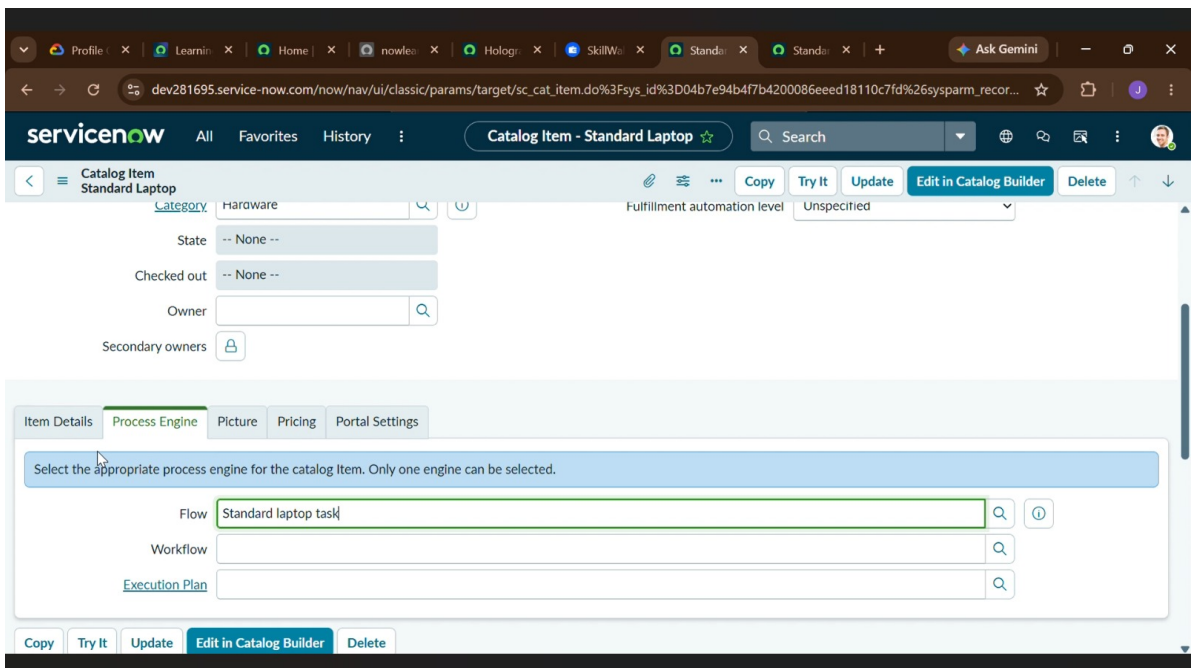


Figure 11.6 — Standard Laptop Process Engine configuration

dev281695.service-now.com/now/nav/ui/classic/params/target/sc_cat_item_list.do%3Fsysparm_first_row%3D1%26sysparm_query%3Dtype!%253Dbun...

servicenow All Favorites History Admin Catalog Items Search

Actions on selected rows... New

All > Type != Bundle > Class != Order guide > Type != Package > Class != Content Item > Published item is empty > Name == standard laptop

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33

Activate Deactivate

1 to 20 of 22

https://dev281695.service-now.com/sc_cat_item.do?sys_id=04b7e94b4f7b4200...

Figure 11.7 — Standard Laptop catalog item

ServiceNow Developers RITM0010004 | Requested IT Operations | Workflow Studio Standard laptop task | Work...

dev281695.service-now.com/now/workflow-studio/page%3FpageType%3Duxf%26url%3D%252Fnow%252Fws-asset%252Fcreate-asset%252Fopsview-65f6768...

Workflow Studio Standard laptop task Flow Global Operations None

Flow executions 2 Last refreshed just now.

Export

Name	State	Runtime	Created by	Created	Sys ID
Standard laptop task	Waiting	187	admin	2026-10-01 02:49:22	53afb9b9b7c2f031011ca98fa5ed323f6
Standard laptop task	Waiting	1,030	admin	2026-10-01 01:54:25	ea1333dbd3ab03106228dfdaf06cd43b

Showing 1-2 of 2 1 10 rows per page

35° ENG IN 3:26 PM 10/1/2026

Figure 11.8 — Flow executions

12. Known Issues

BG-001 is the principal known issue. Task creation can occur before the required approval state. The issue remains Open in the latest UAT record.

13. Future Enhancements

- Add an explicit approval-state condition/wait before task creation.
- Prevent task creation for rejected requests.
- Add requester and Hardware notifications.
- Add richer fulfilment-status visibility.
- Extend the pattern to additional IT hardware items.